

NAME : _____

CLASS : _____

DP014

Physics 1

Semester 1

Session 2024/2025

2 hours

DP014

Fizik 1

Semester 1

Sesi 2024/2025

2 jam



UNIT FIZIK

KOLEJ MATRIKULASI SELANGOR

PRA-PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI 1

MATRICULATION PROGRAMME PRE-EXAMINATION 1

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO

<i>NO SOALAN</i>	<i>MARKAH</i>
1	/8
2	/12
3	/9
4	/12
5	/9
6	/12
7	/18
TOTAL	/80

Kertas soalan ini mengandungi 8 halaman bercetak.

This question paper consists of 8 printed pages

LIST OF SELECTED CONSTANT VALUES

Speed of light in vacuum	c	= $3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space	μ_o	= $4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space	ϵ_o	= $8.85 \times 10^{-12} \text{ F m}^{-1}$
Electron charge magnitude	e	= $1.60 \times 10^{-19} \text{ C}$
Planck constant	h	= $6.63 \times 10^{-34} \text{ J s}$
Electron mass	m_e	= $9.11 \times 10^{-31} \text{ kg}$ = $5.49 \times 10^{-4} \text{ u}$
Neutron mass	m_n	= $1.674 \times 10^{-27} \text{ kg}$ = 1.008665 u
Proton mass	m_p	= $1.672 \times 10^{-27} \text{ kg}$ = 1.007277 u
Deuteron mass	m_d	= $3.34 \times 10^{-27} \text{ kg}$ = 2.014102 u
Molar gas constant	R	= $8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Avogadro constant	N_A	= $6.02 \times 10^{23} \text{ mol}^{-1}$
Boltzmann constant	k	= $1.38 \times 10^{-23} \text{ J K}^{-1}$
Gravitational constant	G	= $6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Free-fall acceleration	g	= 9.81 m s^{-2}
Atmospheric pressure	atm	= $1.013 \times 10^5 \text{ Pa}$

LIST OF SELECTED FORMULAE

1. $v = u + at$
2. $s = ut + \frac{1}{2}at^2$
3. $v^2 = u^2 + 2as$
4. $s = \frac{1}{2}(u + v)t$
5. $v = u - gt$
6. $s = ut - \frac{1}{2}gt^2$
7. $v^2 = u^2 - 2gs$
8. $p = mv$
9. $J = F\Delta t$
10. $J = \Delta p = mv - mu$
11. $f_s \leq \mu_s N$
12. $f_k = \mu_k N$
13. $W = F \cdot s = Fs \cos \theta$
14. $K = \frac{1}{2}mv^2$
15. $U = mgh$
16. $U_s = \frac{1}{2}kx^2 = \frac{1}{2}Fx$
17. $a_c = \frac{v^2}{r} = r\omega^2 = v\omega$
18. $F_c = \frac{mv^2}{r} = mv\omega = mr\omega^2$
19. $s = r\theta$
20. $v = r\omega$
21. $a_t = r\alpha$
22. $\omega = \omega_o + \alpha t$
23. $\theta = \omega_o t + \frac{1}{2}\alpha t^2$
24. $\omega^2 = \omega_o^2 + 2\alpha\theta$
25. $\theta = \frac{1}{2}(\omega_o + \omega)t$
26. $\tau = rF \sin \theta$
27. $\frac{Q}{t} = -kA \left(\frac{\Delta T}{L} \right)$
28. $v_{rms} = \sqrt{\langle v^2 \rangle}$
29. $v_{rms} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}$
30. $Q = \Delta U + W$

Answer **all** questions.

- 1 (a) The mass of a ball is 7.2 g. If the ball has a radius 3.5 cm, find the density of the ball?

[4 marks]

- (b)

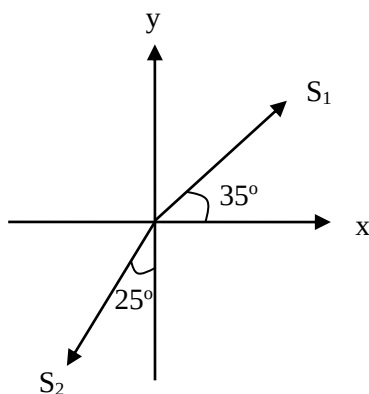


FIGURE 1

FIGURE 1 shows the displacement $S_1 = 5.1$ m and $S_2 = 7.5$ m. Calculate the magnitude and direction of $S = S_1 + S_2$.

[4 marks]

- 2 (a)

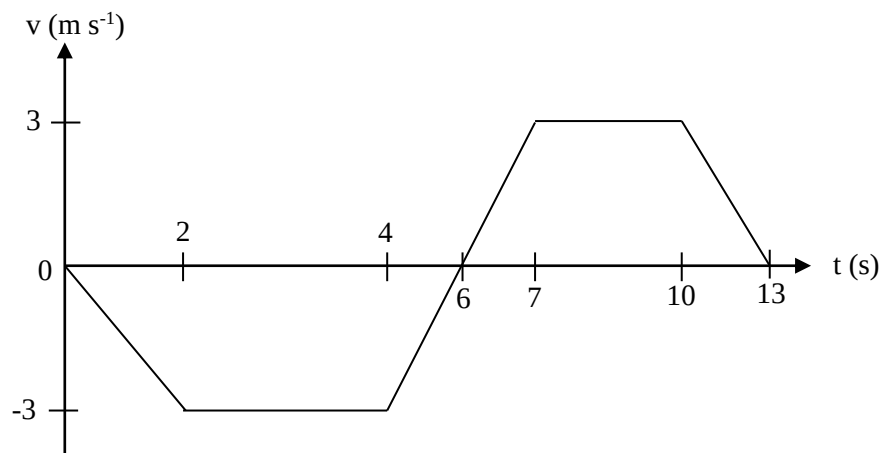


FIGURE 2

FIGURE 2 shows the *velocity-time* graph for the motion of an object.

- (i) Determine the distance of the object from its starting point.
- (ii) Calculate the displacement of the object after 13 s.
- (iii) Calculate the magnitude of the maximum acceleration of the object.

[7 marks]

(c) A stone dropped from a bridge strikes the water after 8 s. Calculate the

- (i) speed just before it strikes the water surface.
- (ii) height of the bridge from the water level.

[5 marks]

3

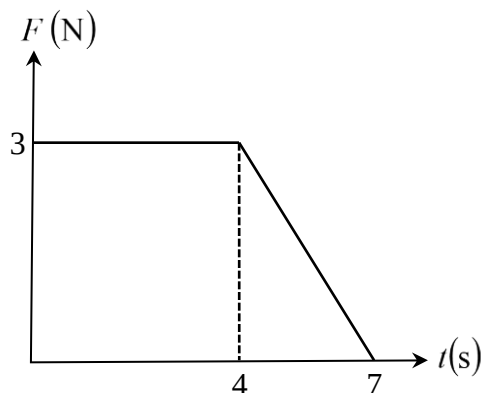


FIGURE 3

FIGURE 3 shows a force - time graph which is acting on 2.5 kg object. Calculate.

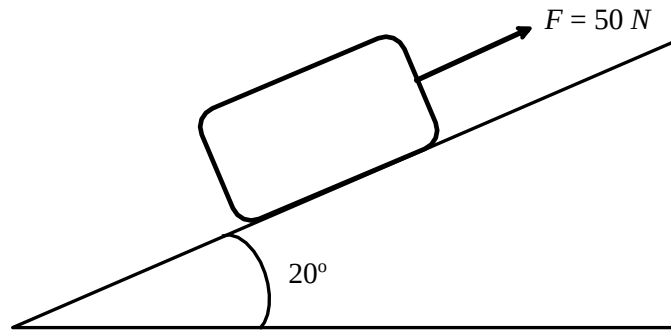
- (i) the impulse of the force.
- (ii) the change in momentum.
- (iii) the final velocity of the object if it is initially at rest.

[5 marks]

- (b) An object $m_A = 6.5 \text{ kg}$ is moving at $u_A = 4.0 \text{ m s}^{-1}$ along x – direction collides head – on with another object $m_B = 11.0 \text{ kg}$ that is moving at $u_B = 5.5 \text{ m s}^{-1}$. They stick together after the collision. Determine the magnitude and direction of the final velocity after the collision.

[4 marks]

4

**FIGURE 4**

1.4 kg box slides up an inclined plane as shown in **FIGURE 4**. The kinetic friction coefficient between the box and the plane is 0.23.

- (a) Sketch **and** label a free body diagram of forces acting on the box.
- (b) Calculate
 - (i) the normal force on the box.
 - (ii) the kinetic frictional force acting on the box.
 - (iii) the net force acting on the box.
 - (iv) the acceleration of the box.

[12 marks]

- 5 A motorcycle moving at 50 km h^{-1} tries to round a corner in a circular arc of 25.0 m radius. The roadway is flat.

- (i) Sketch a free body diagram of the motorcycle.
- (ii) If the distance covered by the motorcycle is 130 m, what angle in degrees has the motorcycle travelled around the corner?
- (iii) How large must the coefficient of friction between the wheels and the roadway if the motorcycle is not to skid?

[9 marks]

- 6 (a) When a ceiling fan is switched on, it rotates from rest in 8 s attains an angular velocity of 50 revolutions per minute. Calculate;
- angular acceleration.
 - the number of revolutions its makes before attaining its final angular velocity.

[7 marks]

(b)

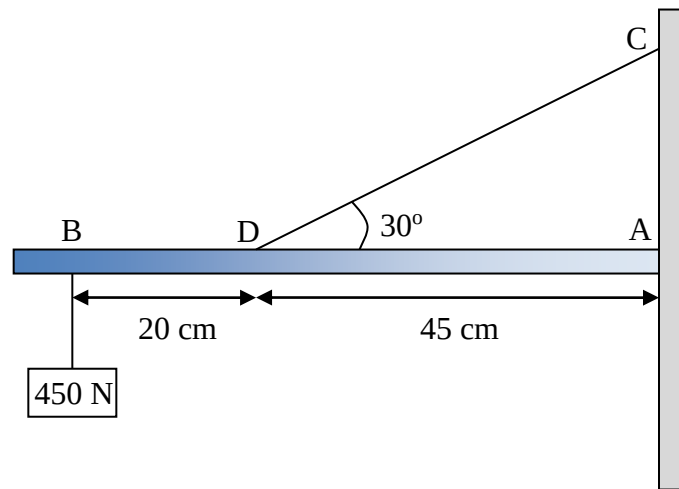


FIGURE 5

A 180 N uniform rod with length 80 cm is hinged to a wall at A. It is held horizontally by a string CD with a 450 N load is suspended at B as shown in the **FIGURE 5** above. The tension in string CD is equal to 1300 N. If the point A is taken as an axis of rotation, calculate

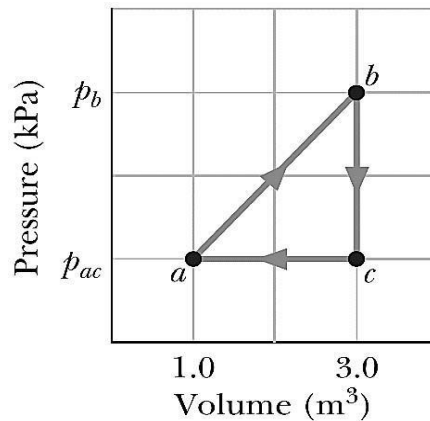
- the torque that rotates in clockwise AND anticlockwise direction?
- the total torque of the system.

[5 marks]

- 7 (a) A closed box is filled with dry ice at temperature -75.5°C , while the outside temperature is 20°C . The box is cubical, measuring 0.40 m on a side, and thickness of the wall is $2.25 \times 10^{-2}\text{ m}$. In one day, $5.5 \times 10^6\text{ J}$ of heat is conducted through the six walls. Determine the thermal conductivity of the material from which the box is made.

[4 marks]

(b)

**FIGURE 6**

A sample of an ideal gas is taken through the cyclic process $abca$ shown in **FIGURE 6**. The scale of the vertical axis is set by $P_b = 7.0$ kPa and $P_{ac} = 1.5$ kPa. At point a , $T = 230$ K. Calculate;

- (i) the number of molecules of gas in the sample,
- (ii) the temperature of the gas at point b ,
- (iii) the temperature of the gas at point c .
- (iv) What is the type of thermodynamic process for part bc **and** ca ?

[11 marks]

- (c) An ideal gas expands adiabatically to three times its original volume. In doing so, the gas does 700 J of work.

- (i) How much heat flows from the gas?
- (ii) What is the change in internal energy of the gas?

[3 marks]

END OF QUESTIONS PAPER